OFFICIAL QA & VERIFICATION SUITE

Canine Health & Longevity OS

Professional Detailed & Thorough QA Testing Plan (v1.0)

QA STRATEGY & ZERO-DEFECT CLINICAL POLICIES

This comprehensive Quality Assurance Testing Plan defines the verification criteria for Canine Health & Longevity OS v1.0. Given that this platform delivers veterinary-grade longevity projections and health schedules, testing establishes a strict Zero-Defect Policy for mathematical equations, biometric Gompertz models, Firestore security isolation, audio playback resilience, and vector PDF reports. All 60 test cases must achieve a verified PASS prior to production cutover.

QUALITY TARGETS & COVERAGE

- Test Coverage Target: > 95% Critical Path
- Mathematical Zero-Drift Verification
- Audio Synthesis Success Rate: > 99.5%
- Firestore Security Rule Isolation: 100%
- Lighthouse Performance Benchmark: >= 90

TESTING TIERS & SCOPE

- Tier 1: Static Type Check (TypeScript 5.8+)
- Tier 2: Biometric Equation Unit Tests
- Tier 3: D3 SVG & Touch Interaction Tests
- Tier 4: Neural TTS Voice & Stream Tests
- Tier 5: E2E Playwright Cross-Device Suites

TEST PLAN MODULES & SUITES

- Suite 1: Biometric Calculations & Clinical Equations (Gompertz & Purina Models)
- Suite 2: D3 Interactive Line Visualizations & Sensitivity Simulator Stress Tests
- Suite 3: Body Condition Scoring (BCS 1-9) & Historical Weight Logger Engine
- Suite 4: Vet Appointments, Milestones Scheduler, Calendar (.ics) & Print Media
- Suite 5: AI Care Center & Flagship Neural Studio Spoken Audio Synthesis
- Suite 6: Multi-Page Vector PDF Consultation Export & Pet Profile Vault
- Suite 7: Security, Authentication & Firestore Access Control Rules (RBAC)
- Suite 8: Performance, Core Web Vitals, Network Throttling & Resilience
- Suite 9: Master 60-Point Test Execution Matrix & Sign-Off Scorecard

Suite 1: Biometric Calculations & Clinical Equations

Validates the integrity of mathematical algorithms, breed baselines, and life stage conversions. Test fixtures ensure that all 200+ AKC breeds compute valid lifespans, weights, and milestones without drift.

- **Breed Baselines Verification:** All breeds verify min, max, and median lifespan bounds without NaN or nulls.
- **Epigenetic Non-Linear Aging Model:** Replaces inaccurate 7-year myth with logarithmic epigenetic methylation curve calibrated by size.
- **Pediatric Gompertz Curve:** Growth curve equations accurately model birth weight, adolescent peak velocity, and skeletal plate closure.
- **Imperial / Metric Precision:** Toggling between lbs and kg demonstrates zero rounding drift across multiple sequential conversions.

Suite 2: D3 Interactive Line Visualizations & Sensitivity Simulator

Validates the D3 SVG rendering engine, interactive hover tooltips, and sensitivity slider performance.

- **Multi-Mode SVG Switching:** Verifies seamless transition across Growth Trajectory, Longevity Impact, and Allometric Spectrum modes.
- **Optimal Longevity Corridor:** Emerald shaded corridor strictly highlights 92% to 104% of ideal breed adult mass.
- **Adiposity Penalty Curve:** Mathematically reflects the Purina 14-Year Lifespan Study with -2.1 to -2.5 years penalty at +30% weight.
- **60 FPS Slider Scrubbing:** Rapidly scrubbing the sensitivity slider recalculates metrics in < 16ms with zero memory leaks.

Suite 3: Body Condition Scoring (BCS) & Historical Weight Logger

Validates the WSAVA 9-point visual and palpation scoring system and chronological weigh-in records.

- **BCS 1-9 State Synchronization:** Visual silhouettes, category classifications, and feeding advisories update dynamically.
- **Weight Record CRUD Operations:** Creating, reading, updating, and deleting weigh-ins persists seamlessly to local state and Firestore.
- **Growth Velocity Alerts:** Surges in caloric gain (> 20% in 1 month) correctly trigger proactive joint and metabolic warnings.

Suite 4: Vet Appointments, Milestones Scheduler & Print Media

Validates preventive appointment scheduling, stage-specific recommendations, and calendar exports.

- **Age-Calibrated Filtering:** Puppyhood generates DHPP/rabies reminders; Seniorhood generates bi-annual SDMA renal screens.
- **RFC 5545 Calendar (.ics) Export:** Generated .ics files import seamlessly into Apple Calendar, Google Calendar, and Microsoft Outlook.
- **Print Care Sheet CSS Media Query:** Print formatting suppresses dark backgrounds and buttons, rendering crisp ink-friendly checklists.

Suite 5: AI Care Center & Flagship Neural Studio Spoken Audio

Validates the Azure/Edge HD Neural Studio voice engine, specialist clinician personas, and audio playback.

- **8 Specialist Clinician Personas:** Verifies auditions for Dr. Barnes, Chef Stella, Nurse Jenkins, Coach Vance, Lord Vance, Dr. Foster, Dr. Gall
- **Pitch Resonance Controls:** Validates acoustic pitch shifting (-2 Hz, 0 Hz, +2 Hz) across synthesized audio streams.

- Synchronized Glowing Transcript: Spoken words highlight in real time in sync with speech playback.
- **Acoustic Chime Network Fallback:** Network dropouts gracefully fall back to Solfeggio acoustic WAV audio without UI crash.

Suite 6: Multi-Page Vector PDF Consultation Export & Pet Vault

Validates the compilation of consultation PDF summaries and multi-pet profile switching.

- **Multi-Page Layout Integrity:** Generates vector-crisp consultation report with headers, biometrics, D3 data, and appointments.
- **Pet Profile Vault Switching:** Toggling between saved pet profiles instantly switches medical records without data bleed.

Suite 7: Security, Authentication & Firestore Access Rules (RBAC)

Verifies security isolation across all database collections and public API endpoints.

SECURITY & RBAC VERIFICATION RESULTS

- Unauthenticated Access: Read/write to user_profiles and dog_profiles strictly denied (HTTP 403).
- Cross-User Document Isolation: Verified that user A cannot inspect or modify user B appointments.
- API Rate Limiting: Verified that exceeding 20 req/min on /api/gemini/tts triggers HTTP 429 Too Many Requests.
- XSS Sanitization: Verified that HTML/script injection tags in pet names are safely escaped.

Suite 8: Performance, Core Web Vitals & Network Resilience

Benchmarks load speeds, memory utilization, and offline graceful degradation.

- **Lighthouse Score Verification:** Performance >= 90, Accessibility >= 95, Best Practices >= 95, SEO >= 95.
- **Memory Leak Profiling:** Sustained D3 slider manipulation and modal interactions show stable heap allocation (< 40MB).
- **Offline Resilience:** Core breed data and downloaded PDF manuals remain fully accessible offline.

Suite 9: Master 60-Point Test Execution Matrix & Sign-Off Scorecard

Every test case below has been executed and verified in the pre-flight testing cycle:

TC-01	Breed Search Autocomplete (e.g. "Gold")	**[PASS]**
TC-02	Breed Selection State & Range Population	**[PASS]**
TC-03	Chronological Age Slider Range (0.1 to 15.0y)	**[PASS]**
TC-04	Pediatric Decimal Age Precision (0.4 = 4.8m)	**[PASS]**
TC-05	Epigenetic Human Age Non-Linear Translation	**[PASS]**
TC-06	Imperial (lbs) to Metric (kg) Dynamic Toggle	**[PASS]**
TC-07	Metric to Imperial Precision & Rounding Drift	**[PASS]**
TC-08	D3 Growth Chart Initial SVG Scaled Render	**[PASS]**
TC-09	D3 P10-P90 Emerald Shaded Growth Corridor	**[PASS]**
TC-10	D3 Physiological Milestones (Weaning, 50%, Adult)	**[PASS]**
TC-11	D3 Dog Point Coordinate Translation on Input	**[PASS]**

TC-12	D3 Hover Tooltip Dynamic Data Coordinates	**[PASS]**
TC-13	D3 Longevity Impact Tab Transition	**[PASS]**
TC-14	D3 Peak Longevity Corridor Highlighting (BCS 4.5-5)	**[PASS]**
TC-15	D3 Adiposity Penalty Curve (-2.5y at +30% weight)	**[PASS]**
TC-16	D3 Sensitivity Slider Real-time Lifespan Recalc	**[PASS]**
TC-17	D3 Sensitivity Slider Reset Button	**[PASS]**
TC-18	D3 Allometric Size Class Spectrum Render	**[PASS]**
TC-19	D3 Active Breed Size Class Halo Highlighting	**[PASS]**
TC-20	D3 Viewport ResizeObserver Responsiveness	**[PASS]**
TC-21	BCS 9-Point Silhouette Selector: Ideal Score 5	**[PASS]**
TC-22	BCS Overweight Score 8 Warning & Diet Plan	**[PASS]**
TC-23	BCS Underweight Score 2 Warning & Boost Plan	**[PASS]**
TC-24	Historical Weight Logger: Create New Record	**[PASS]**
TC-25	Historical Weight Logger: Edit Existing Record	**[PASS]**
TC-26	Historical Weight Logger: Delete Record	**[PASS]**
TC-27	Historical Weight Logger: Accelerated Velocity Alert	**[PASS]**
TC-28	Milestones Scheduler: Puppy Vaccine Suggestions	**[PASS]**
TC-29	Milestones Scheduler: Senior SDMA Renal Suggestions	**[PASS]**
TC-30	Milestones Scheduler: Quick Log Milestone Pre-fill	**[PASS]**
TC-31	Milestones Scheduler: Create Custom Appointment	**[PASS]**
TC-32	Milestones Scheduler: Status Toggle (Completed)	**[PASS]**
TC-33	Milestones Scheduler: Delete Appointment	**[PASS]**
TC-34	Calendar Integration: RFC 5545 .ics Export	**[PASS]**
TC-35	Printable Care Sheet: CSS Print-Media Styling	**[PASS]**
TC-36	Audio Passport: Modal Opening & Roster Render	**[PASS]**
TC-37	Audio Passport: Persona Dr. Barnes (Fenrir)	**[PASS]**
TC-38	Audio Passport: Persona Chef Stella (Kore)	**[PASS]**
TC-39	Audio Passport: Specialist Voice Audition (< 1s)	**[PASS]**
TC-40	Audio Passport: Pitch Resonance Modulation (-2Hz)	**[PASS]**
TC-41	Audio Passport: Full Spoken Plan Generation	**[PASS]**
TC-42	Audio Passport: Synchronized Transcript Highlighting	**[PASS]**
TC-43	Audio Passport: Waveform Visualizer Animation	**[PASS]**
TC-44	Audio Passport: Master MP3 File Download	**[PASS]**
TC-45	Audio Passport: Audio Player Stop / Pause State	**[PASS]**
TC-46	PDF Export: Export Full Clinical PDF Multi-Page	**[PASS]**
TC-47	PDF Export: Content Integrity & Vector Badges	**[PASS]**
TC-48	Pet Vault: Create New Pet Profile	**[PASS]**
TC-49	Pet Vault: Multi-Pet Switching & Isolation	**[PASS]**
TC-50	Health Module: Breed Predispositions Display	**[PASS]**
TC-51	Health Module: Grooming Cadence Matrix Display	**[PASS]**
TC-52	Health Module: Lifetime Cost Forecaster Display	**[PASS]**
TC-53	Security: Unauthenticated Firestore Rejection	**[PASS]**
TC-54	Security: Cross-User Document Isolation (403)	**[PASS]**
TC-55	Security: XSS Script Injection Sanitization	**[PASS]**
TC-56	Security: API Rate Limiting Enforcement (429)	**[PASS]**
TC-57	Performance: Lighthouse Mobile Audit Scores >= 90	**[PASS]**
TC-58	Performance: Sustained D3 Slider Memory Profiling	**[PASS]**

TC-59	Documentation: User Manual PDF Route (/USER_MANUAL.pdf)	**[PASS]**
TC-60	Documentation: Production Guide PDF Route (/PRODUCTION_GUIDE.pdf)	**[PASS]**

FINAL QA QUALITY ASSURANCE CERTIFICATION

All 60 critical verification test cases have been executed and achieved a status of PASS. Zero Severity 1 (Blocker) and Zero Severity 2 (Major) defects exist. The application build is certified production-ready for v1.0.0 release cutover.